

	September	October	November	December	January	February	March	April	May	June
6th Grade	Trimester 1			Trimester 2			Trimester 3			
	Variables and Design - 3-Year Car Challenge Theme Connection - Power of one variable Diversity of Life Theme Connection - Single-cell life			Weather and Water Theme Connection: Effect of one atmospheric condition			Human Systems Interactions Theme Connection: Importance of one body system project Variables and Design Theme Connection - Power of one variable			
	Variables and Design - 3 Year Car Challenge			Weather and Water			Human Systems Interactions			
	Investigation 1: Independent vs Dependent Variables Part 1: Effect of ramp angle on the same car. Part 2: Graphing scientific data Part 3: Engineering design process - building a gravity powered car Part 4: Comparing designs by distance at the same ramp angle			Homework Project - Long-term weather monitoring project			Major Class Project - Body System Poster			
	Diversity of Life Homework Project - Cell Cake Model			Investigation 1: What is Weather? Part 1: Intro to the Weather - What is weather? Part 2: The Air Around Us - What is air? Part 3: Earth's Atmosphere - What is the atmosphere?			Investigation 1: Systems Connections Part 1: Human Body Structural Levels - What is a human body made of? Part 2: Systems Research - How do human organ systems interact? System Spotlight: Digestive System Spotlight: Circulatory			
	Investigation 1: What is Life? Part 1: Living or Nonliving - How do you know if something is living? Part 2: Is Anything Alive in Here? - How do you know if something is living?			Investigation 3: Air Pressure and Wind Part 1: Air-Pressure Inquiry - How does pressure affect air?			Investigation 2: The Nervous System Part 1: The Sense of Touch - How does the sense of touch work in humans? Part 2: Sending a Message - How do messages travel to and from the brain? Part 3: Other Senses - How are the senses alike and how are they different? Part 4: Learning and memory - How do humans learn and form memories?			
	Investigation 2: The Microscope Part 1: Meet the Microscope - How do objects appear when they are viewed			Investigation 3: Convection Part 1: Density of Fluids - What is the relationship between layering of fluids and density? Part 2: Convection in Water - How does heat affect density of fluids? Part 3: Convection in Air - How do gases flow in the atmosphere?			Investigation 3: Supporting Cells Part 1: Food and Oxygen - How do cells in the human body get the resources they need? Part 2: Aerobic Cellular Respiration - How does the energy in food become energy that cells			
	Part 2: Field of View - How can we estimate the size of an object by looking at it			Investigation 4: Radiation Part 1: Latitude - How does the weather differ between locations? Part 2: Solar Angle - How does the sun affect the temperature of locations on Earth? Part 3: Heating Earth - What factors affect the surface temperature of Earth?			Variables and Design Investigation 1: Testing Variables Part 1: Air Trolleys - How can we describe and measure motion in a system? Part 2: Controlled Experiment - What variables affect the operation of an air trolley? Part 3: Design an Experiment - Student-generated question regarding air trolley.			
	Part 3: Microscopic Life - What evidence can we fins that brine shrimp are a living			Investigation 5: Conduction Part 1: Fluid Conduction - How does energy move through materials? Part 2: Insulation - Student-generated question about energy transfer. Part 3: Home Design - How can we design a more efficient way to decrease energy			Investigation 2: Testing Designs Part 1: Air-Trolley Design Challenge - How do engineers decide what to change in a design? Part 2: Engineering Design Cycle - What are the steps taken to solve a problem in engineering?			
	Investigation 3: The Cell Part 1: Discovering Cells - What microscopic structures make up organisms such			Investigation 6: Air Flow Part 1: Atmospheric Heating - How does the atmosphere heat up? Part 2: Local Winds - How does the energy from the Sun affect wind on Earth? Part 3: Global Winds - What affects the direction of global winds?			Investigation 3: Real-World Problems Part 1: Define a Problem - How does understanding variables help us define a problem and Part 2: Future Tech - How does technology help engineers solve problems?			
	Part 2: Paramecia - How are elodea and the paramecium alike, and how are they			Investigation 7: Water in the Air Part 1: Is Water Really There? - Student-generated question about water vapor. Part 2: Phase Change and Energy Transfer - How does energy transfer when water transfer			Additional Engineering Projects if there's time.			
	Part 3: Microworlds - Is there life in the minihabitats? If so, where did it come Part 4: Human Cheek Tissue - What microscopic structures make up organic			Investigation 8: The Water Planet Part 1: Water-Cycle Simulation - What is the water cycle? Part 2: Ocean Currents - What affects the direction that ocean water flows? Part 3: Ocean Climate - How does the ocean affect climate on land? changes phases?						
	Investigation 4: Domains Part 1: Comparing Living Things - What are the building blocks of cell structures? Part 2: Bacteria - What evidence is there that bacteria is a living organism? Part 3: Fungi - What evidence is there that fungi are living organisms? Part 4: Archaea: The Three Domains - What are the characteristics of archaea?			Investigation 9: Climate Over Time Part 1: Climate Change - How have climates changed over time? Part 2: The Role of Carbon Dioxide - How do greenhouse gasses in the atmosphere						
	Investigation 5: Plants: The Vascular System Part 1: What Happened to the Water? - Student-created question, e.g., 'What			Investigation 10: Meteorology Part 1: Weather Maps - What information can you get from a weather map? Part 2: Identify Key Ideas - What makes weather happen?						
	Part 2: Looking at Plant Structures - How does water travel through a plant? Part 3: Transpiration and Photosynthesis - How do plants use water?			Part 3: Climate in the News - What are the effects of a slight rise in global temperature?						
	Investigation 6: Lima Bean Dissection Part 1: How do the structural adaptations of seeds help them survive? Part 2: Environmental and Genetic Factors - How do environmental factors affect									
	Part 3: Flowering-Plant Reproduction - What is the role of a flower? Part 4: Flowers and Pollinators - What adaptations do flowering plants have to									
	Investigation 7: Variation of Traits Part 1: Inheriting Traits - Ho do traits pass from parents to offspring? Part 2: Modeling Heredity - How does sexual reproduction produce variation in									
	Investigation 8: Insects Part 1: Structure, Function, and Behavior - How do the structures and behaviors									
	Part 2: Insect Systems - How is the insect transport system like plant and human									
	Investigation 9: Diversity of Life Part 1: Bioblitz - What kind of plant and animal life exists in our schoolyard? Part 2: What is Life - How do you know if something is living?									
	Trimester 1			Trimester 2			Trimester 3			
	Variables and Design - 3-Year Car Challenge Theme Connection - Power of one variable Chemical Interactions Theme Connection - Power of one discovery (microscopes, atoms etc.)			Earth History Theme Connection - Power of one event (volcanism, etc)			Populations and Ecosystems Theme Connection - Keystone species			

7th Grade	Variables and Design - 3 Year Car Challenge Investigation 2: Materials Engineering Investigation Part 1: Engineering design process - building a balloon powered car Part 2: Effect of materials on distance traveled Part 3: Graphing scientific data Part 4: Comparing designs and analyzing data Chemical Interactions Major Class Project - Build a freezer Investigation 1: Substances Part 1: Mystery Mixture - How can we find out what the two substances are in the Part 2: Mixing Substances - How can we find out what two substances are in the Investigation 2: Elements Part 1: Periodic Table - What is the periodic table if the elements? Part 2: Elements in the World - What makes up all the substances on Earth? Investigation 3: Particles Part 1: Capture the Gas - How can the gas produced in a chemical reaction be Part 2: Air is Matter - Is air matter? Does air have mass and take up space? Part 3: Air as Particles - What is the relationship between particles in matter? Investigation 4: Kinetic Energy Part 1: Gas Expansion/Contraction - What happens to particles in a sample of air Part 2: Liquid Expansion/Contraction - What happens to particles in a sample of Part 3: Solid expansion/contraction - What happens to particles in a sample of Investigation 5: Energy Transfer Part 1: Mixing Hot and Cold - If two equal volumes of hot and cold water are Part 2: Particle collisions - How does the energy transfer from one substance to Part 3: Heat - How is heat measured? Investigation 6: Thermos Engineering Part 1: Insulation - Student-generated question regarding energy transfer. Part 2: Thermos Design - What is the best thermos design? Investigation 7: Solutions Part 1: Dissolve and Melt - What is the difference between dissolving and Part 2: Solubility - Do all substances form solutions in water? Investigation 8: Phase Change Part 1: Melting Temperature - What happens at the particle level when a Part 2: Adding Thermal Energy - What is the relationship between melting and Part 3: Freezing Water - How can you freeze water in the classroom? Part 4: Changing Phase - What are all the ways that a substance can change Investigation 9: Reaction Part 1: Substance Models - How do atoms combine to make new substances? Part 2: Limewater Reaction - What happens at the particle level during a chemical Part 3: Baking Soda and Acid - What is the chemical reaction between Investigation 10: Limiting Factors Part 1: Citric Acid and Baking Soda - What is a limiting factor in a chemical Part 2: Identifying Key Ideas - What have I learned about chemical interactions?	Earth History Major Project TBD Investigation 1: Earth is Rock Part 1: What's the Story of This Place? - Which landforms occur at different locations on Earth? Part 2: Grand Canyon Rocks - Why do there appear to be stripes on the walls of the Grand Part 3: Correlating Grand Canyon Rocks - Why do there appear to be stripes on the walls of Investigation 2: Weathering and Erosion Part 1: Stream Table - What happens to Earth materials when water flows over landforms? Part 2: Weathering - How did weathering and erosion contribute to the formation of the Grand Part 3: Soils - How is soil related to rocks? Investigation 3: Deposition Part 1: Sandstone and Shale - What happens to sediments that get deposited in basins? Part 2: Limestone - How does limestone form? Part 3: Interpreting Sedimentary Layers - What do sedimentary layers reveal about ancient Investigation 4: Fossils and Past Environments Part 1: Fossils - How do fossils get in rocks? Part 2: A Long Time Ago - How old are fossils? Part 3: Index Fossils - When did the Grand Canyon Rocks form? Investigation 5: Igneous Rock Part 1: Earth's Layers - How do igneous rocks form? Part 2: Salol Crystals - Student-generated question regarding crystals. Part 3: Types of Igneous Rocks - What can crystal size tell us about where an igneous rock Investigation 6: Volcanoes and Earthquakes Part 1: Mapping Volcanoes and Earthquakes - Where do volcanoes occur on Earth? Where do Part 2: Moving Continents - Why do volcanoes and earthquakes occur where they do? Part 3: Plate Tectonics - What causes plates to move? Investigation 7: Mountains and Metamorphic Rocks Part 1: Plate Models - What happens to Earth's crust during plate interactions? Part 2: Metamorphic Rocks - How do metamorphic rocks form? Investigation 8: Geoscenarios Part 1: Introduction to the Project - What do we need to know to tell the geologic story of a Part 2: Research and Writing - What do we need to know to tell the geographic story of a place? Part 3: Presentations - What do we need to know to tell the geographic story of a place? Investigation 9: What is Earth's Story? Part 1: Revisit the Grand Canyon - What is the geologic story of the Grand Canyon? Part 2: Review the Evidence - How do Earth materials cycle through constructive and	Populations and Ecosystems Homework Project - Ecosystem Model Investigation 1: Milkweed Bugs Part 1: Introducing Milkweed Bugs - What does a population of milkweed bugs need to survive in Part 2: Milkweed-Bug Habitat - What needs to be considered when building a habitat for Part 3: Observing Milkweed Bug Habitats - How do milkweed bugs reproduce and grow? Investigation 2: Sorting out Life Part 1: Ecosystem Card Sort - What is the relationship between individuals, populations, Part 2: Video Population Study - How is the milkweed-bug-habitat study similar to and different Part 3: Ecoscenarios - Student-generated question about ecosystems. Investigation 3: Mono Lake Part 1: A visit to Mono Lake - What are the different biotic and abiotic components of the Mono Part 2: Mono Lake Food Web - How do the organisms at Mono Lake interact? Part 3: Ecoscenario Food Webs - How do the organisms in your ecoscenario interact? Investigation 4: Minihabitats Part 1: The Physical Environment - What abiotic factors should be considered when setting up Part 2: Introducing Life - What interactions are likely for the organisms in the mini habitat? Part 3: Observing Minihabitats - What interactions and changes have taken place in the terariums Investigation 5: Producers Part 1: Growing Producers - What is the effect of light on producers? Part 2: Biomass and Producers - How do producers grow and increase biomass? Part 3: Ecoscenario Producers - What are the roles of specific producers in the ecosystem? Part 4: Energy Transfer From Food - How can we model and measure energy transfer from food? Investigation 6: Following the Energy Part 1: Using Energy - What are the kinds of work you do that require energy? Part 2: Food Chain Game - What is needed to sustain a food chain? Part 3: Trophic Levels - How do biomass and energy flow through an ecosystem? Part 4: Decomposers - What happens to the energy stored in the biomass of an organism when it Investigation 7: Population Size Part 1: Reproductive Potential - What factors affect how many milkweed bugs could be in your Part 2: Limiting Factors - What limiting factors affect algae and brine shrimp populations at Mono Part 3: Population Dynamics - How does predicted population growth compare to actual Investigation 8 - Human Impact Part 1: Biodiversity - Why is biodiversity important in an ecosystem? Part 2: Invasive Species - What can happen when a species is introduced to an ecosystem? Part 3: Mono Lake Revisited - What impact have people had on Mono Lake? Investigation 9: Ecoscenarios Part 1: Human Involvement - How have humans affected your ecoscenario, and what efforts have Part 2: Evaluating Solutions - How have humans affected your ecoscenario, and what efforts Part 3: Presentations - How have humans affected your ecoscenario, and what efforts have
	Trimester 1 Variables and Design - 3-Year Car Challenge Theme Connection - Power of one variable Electromagnetic Force Theme Connection - Changes in one force Waves	Trimester 2 Waves Continued Theme Connection - Power of one discovery Heredity and Adaptation Theme Connection - Impact of one discovery Planetary Science	Trimester 3 Planetary Science Continued Theme Connection - Impacts of individual scientists Gravity and Kinetic Energy Theme Connection - Power of one's own ideas

	Theme Connection - Power of one discovery	Theme Connection - Impacts of individual scientists	
8th Grade	Variables and Design - 3 Year Car Challenge	Waves Continued	Planetary Science Continued
	Investigation 2: Velocity, Acceleration, and Forces Part 1: Speed, Velocity, and Acceleration Part 1: Engineering design process - building an electric motor-powered car Part 3: Graphing scientific data Part 4: Comparing designs and analyzing data	Investigation 3: Light Waves Part 1: Mirrors - What happens when light waves interact with matter? Part 2: Spectra - What do spectra reveal about light? Part 3: Color - Student-generated question, e.g. 'What makes objects appear as different colors' Part 4: Refraction - What happens to light waves at the interface between different media?	Investigation 3: Moon Study Part 1: A Close Look at the Moon - What is visible on the Moon? Part 2: How Big/Far? - What does an Earth/Moon scale model look like?
	Electromagnetic Force Major Project - Build a light-up pinball machine	Investigation 4: Communication Waves Part 1: Optical Fibers - What are some design constraints in fiber-optic communications? Part 2: Sending Sound - How is sound sent through radio waves? Part 3: Sending Images - How are images sent through radio waves?	Investigation 4: Phases of the Moon Part 1: Observed Patterns - What Moon-phase patterns can be observed? Part 2: Moon-Phase Models - What causes Moon phases? Part 3: Moon-Phase Simulation - How do models help us understand phases of the moon?
	Investigation 1: What is Force? Part 1: Push and Pull - What makes things move? Part 2: Friction - How does friction affect the force needed to move an object? Part 3: Forces in Action - How do multiple forces affect motion?	Heredity and Adaptation	Investigation 5: Craters Part 1: Moon Craters - Are moon craters the result of volcanoes or impacts? Part 2: Target Earth - Will Earth experience a major impact in the future?
	Investigation 2: The Force of Magnetism Part 1: Properties of Magnets - What happens when magnets interact? Part 2: Magnetic Fields - How can we detect a magnetic field? Part 3: Force over Distance - What factors affect the force of attraction between	Investigation 1: The History of Life Part 1: The Fossil Record - What does the fossil record tell us about the history of life on Earth? Part 2: Transitions - What does the fossil record tell us about how life has changed over time?	Investigation 6: Beyond the Moon Part 1: What's Out There? - What is in the solar system? Part 2: Origins - Where did the solar system come from?
	Investigation 3: Electromagnetism Part 1: Building a circuit - What is required to complete an electric circuit? Part 2: Building an Electromagnet - How does an electromagnet work? Part 3: Improving the Design - Student-generated-question, e.g. How does ____	Investigation 2: Heredity Part 1: Lines of Descent - How can a model help us understand the relationships among Part 2: Inheriting Traits - What leads to variation in a population? Part 3: Modeling Heredity - How can we model how genetic information passes from generation Part 4: Punnett Squares: How can we predict the distribution of traits in a future generation?	Investigation 7: The Solar System Part 1: Where are the Planets? - Where are the planets in the solar system? Part 2: Comparing Temperatures and Atmospheres - Which planet is most like Earth? Part 3: Where is the Water? - Where is the water in the solar system? Part 4: Changing Systems - What impact do humans have on Earth's Systems?
	Waves		Investigation 8: Space Exploration Part 1: Light Spectra - Why is light important in astronomy? Part 2: Exploration of the Solar System - What are the big questions that guide space
	Investigation 1: Make Waves Part 1: Pulse Rate - What is frequency Part 2: Spring Waves - What defines a wave?	Investigation 3: Evolution Part 1 - Adaptation - How do genetic mutations lead to variations in a population? Part 2 - Natural Selection - How do populations change over time? Part 3 - Genetic Technology - How are humans influencing inheritance?	Investigation 9: Orbits and New Worlds Part 1: The Moons of Jupiter - What can be learned by studying the moons of Jupiter? Part 2: Looking for Planets - How are exoplanets found? Part 3: What is our Cosmic Address - Where are you when you are in science class?
	Investigation 2: Wave Energy Part 1: Energy in Waves - What is the relationship between wave properties and Part 2: Bridge Collapse - How are engineering challenges solved? Part 3: Energy in Sound Waves - What is the best way to insulate a recording	Planetary Science Homework Project: Solar System Model	8th Grade Field Trip Evergreen and Beach (?)
		Investigation 1: Earth as a System Part 1: School to Space - Where are you when you are in science class? Part 2: Earth's Systems - Why is Earth Described as a System? Part 3 - Moon Watch - How does the moon change day by day?	Gravity and Kinetic Energy + Paper Coasters
		Investigation 2: Earth/Sun Relationship Part 1: Day and Night - What causes day and night? Part 2: Summer Heat - Why is it hotter in summer? Part 3: Day Length - Why are there more hours of sunlight in the summer?	Investigation 1: Acceleration Part 1: Speed Tracks - What is speed? Part 2: Acceleration Track - What is acceleration? Part 3: Acceleration of Gravity - What is gravity?
		Investigation 1: Earth as a System Part 1: School to Space - Where are you when you are in science class? Part 2: Earth's Systems - Why is Earth Described as a System? Part 3 - Moon Watch - How does the moon change day by day?	Investigation 2: Force of gravity Part 1: Mass and Weight - What is the relationship between mass and weight? Part 2: How Heavy? - What is gravity like on other planets compared to Earth?
		Investigation 2: Earth/Sun Relationship Part 1: Day and Night - What causes day and night? Part 2: Summer Heat - Why is it hotter in summer? Part 3: Day Length - Why are there more hours of sunlight in the summer?	Investigation 3 - Energy and Collisions Part 1: Potential and Kinetic Energy - How is potential energy related to kinetic energy? Part 2: Stop or Crash - How does the kinetic energy of an object change when its speed or mass Part 3: Marble Collisions - How do Newton's laws help us explain marble billiards?
			Investigation 4: Collision Engineering Part 1: Helmet Design Challenge - Which Properties of physics can help us design Part 2: Big ideas - How can we explain the motion of objects?
			Paper Roller Coasters - How can we use the principles of motion to create an exciting roller